

Problem Landscape and Success Criteria for One-User Offline-to-Online Personalized Policies

Research Position and Evaluation Contract

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1 Setting

Evidence base. Two project PRISMA runs yielded 6,663 raw records and 119 full-text reviews. We cite primary papers for their methods and evaluation settings, not as evidence that our policy works.

One user is the unit of personalization. For user u , the offline history

$$H_u^{(0)} = \{\text{thoughts, conversations, corrections, preferences, task states, actions, tools, and outcomes}\}$$

produces an initial checkpoint $c_u^{(0)}$. After deployment, interaction batch $D_u^{(t)}$ is generated by the active checkpoint $c_u^{(t)}$. The update is

$$c_u^{(t+1)} = \mathcal{A}(c_u^{(t)}, H_u^{(0)}, D_u^{(1:t)}) . \quad (1)$$

Each record stores its user, time, task, checkpoint, trace, feedback source, and training eligibility. A shared prior is allowed; raw histories remain separate.

A policy update is required. Retrieval, profiles, memory, and decoding control are baselines, not the learned object. OPPU trains a private user adapter [Tan et al., 2024]; P-RLHF trains a user-conditioned policy [Li et al., 2024]; and SPRInG updates a user adapter over time [Kim and Kim, 2026]. CoSteer is tuning-free [Lv et al., 2025], while FABLE learns an external execution policy around a frozen agent [Jin et al., 2026]. Neither CoSteer nor FABLE updates a deployed user’s language-model policy weights.

Objective. Improve personalization without degrading task success or safety:

$$\begin{aligned} \max_{\mathcal{A}} \quad & \mathbb{E}[U_{\text{pers}}(c_u^{(t+1)})] \\ \text{s.t.} \quad & \mathbb{E}[U_{\text{task}}(c_u^{(t+1)}) - U_{\text{task}}(c_u^{(t)})] \geq -\delta_{\text{task}}, \\ & \mathbb{E}[U_{\text{safe}}(c_u^{(t+1)}) - U_{\text{safe}}(c_u^{(t)})] \geq -\delta_{\text{safe}}, \\ & C_{\text{train}}, C_{\text{serve}}, C_{\text{feedback}} \text{ remain within declared budgets.} \end{aligned} \quad (2)$$

Exact scope. The target is one persistent user policy, initialized from that user’s history, updated from later same-user interactions, used for agent tasks, and trained through language-model parameter updates. We call an update *online* only when post-deployment evidence changes model parameters; fresh rollouts within one training job do not qualify. No cited method covers all five properties.

2 Related Problems and Setting Gaps

2.1 Learning the user

P1: identify preferences from sparse, mixed evidence. A user supplies requests, choices, edits, and task outcomes, not one uniform label. Population training can blur minority preferences [Li et al., 2024, Jang et al., 2023]. CoPL and PROPER share structure across users [Choi et al., 2025, Zhang et al., 2025]; FSPO adapts from few examples [Singh et al., 2025]. We instead use one user’s full history.

Prior-work setting. **One user:** represented, but several methods train across users; offline initialization: yes; post-deployment update: no; personalized agent: no; LLM-policy training: P-RLHF, PROPER, and FSPO, but not CoPL. **Difference:** fixed-data preference learning, not deployed agent updates.

P2: separate persistent preference from transient context. A request may express a lasting preference, a task constraint, or noise. SPRInG filters for drift before updating a user adapter [Kim and Kim, 2026]. Agent traces add tool and reasoning failures that are not preference evidence.

Prior-work setting. **One user:** yes; offline initialization: yes; post-deployment update: no; personalized agent: no; LLM-policy training: yes. **Difference:** SPRInG replays a fixed chronological corpus rather than live agent trajectories.

P3: handle cold start without losing ownership. Collaborative and federated models improve sparse-user adaptation [Choi et al., 2025, Zhang et al., 2025, Zhu et al., 2026]. Our setting may use a shared prior, but histories, updates, and evaluation remain user-specific.

Prior-work setting. **One user:** represented through shared training; offline initialization: yes; post-deployment update: no; personalized agent: no; LLM-policy training: PROPER and FedPDPO, while CoPL trains a reward model. **Difference:** these methods pool users offline; our known-user setting does not require cold start.

2.2 Moving from offline history to online learning

P4: build an offline initializer with online headroom. The offline checkpoint must complete tasks yet remain adaptable. OPPU provides a per-user initializer [Tan et al., 2024]; WebRL shows the value of warm starts for sparse-reward agents [Qi et al., 2025].

Prior-work setting. **One user:** OPPU only; offline initialization: yes; post-deployment update: no; personalized agent: WebRL only; LLM-policy training: yes. **Difference:** neither paper combines these properties.

P5: cross the offline–online boundary without a performance dip. Offline records come from older policies. Synchronization [Lanchantin et al., 2025], hybrid preference learning [Bose et al., 2024], and policy expansion [Zhang et al., 2023] address this handoff. Our handoff must also preserve user, feedback, and checkpoint lineage.

Prior-work setting. **One user:** no; offline initialization: yes; post-deployment update: no; personalized agent: no; LLM-policy training: Bridging only. **Difference:** their “online” data are collected during training, not from a deployed user.

P6: adapt to drift without forgetting or chasing noise. SPRInG selects likely preference shifts and retains residual memory [Kim and Kim, 2026]. Stable Asynchrony handles stale trajectories from lagged policies [Huang et al., 2026]. We need both without forgetting stable preferences.

Prior-work setting. **One user:** SPRInG only; offline initialization: SPRInG only; post-deployment update: no; personalized agent: no; LLM-policy training: yes. **Difference:** neither joins preference drift and stale agent trajectories after deployment.

P7: learn quickly while minimizing user burden. User feedback is expensive. Hybrid preference optimization trades offline data against online queries [Bose et al., 2024]. FABLE learns a pure-online per-user execution policy [Jin et al., 2026]. We measure interactions needed to reach a fixed future-task utility.

Prior-work setting. **One user:** FABLE only; offline initialization: hybrid preference optimization only; post-deployment update: FABLE only; personalized agent: FABLE only; LLM-policy training: not FABLE. **Difference:** FABLE updates an external controller, not an initialized LLM policy.

2.3 Learning an agent policy

P8: preserve task completion while personalizing execution. Personalized agents choose tools, memory, clarifications, and response style while completing the task. PersonalWAB studies web actions [Cai et al., 2025], FingerTip 20K studies mobile execution [Yang et al., 2026], and FABLE adapts execution around a frozen agent [Jin et al., 2026].

Prior-work setting. **One user:** persistent policy only in FABLE; offline initialization: no persistent per-user checkpoint; post-deployment update: FABLE only; personalized agent: yes; LLM-policy training: not across both phases. **Difference:** none trains one personal LLM across both phases.

P9: assign credit from long, partially observed trajectories. A correction may follow many reasoning and tool steps. WebRL learns from sparse terminal feedback [Qi et al., 2025]; we must also identify which steps reveal the user’s preference.

Prior-work setting. **One user:** no; offline initialization: yes; post-deployment update: no; personalized agent: no; LLM-policy training: yes. **Difference:** WebRL’s credit is task-conditioned, not user-conditioned.

2.4 Obtaining credible and affordable evidence

P10: use simulators without training to simulator artifacts. UserRL, UserVille, ProPerSim, and CollabLLM train or evaluate agents with simulated users [Qian et al., 2025, Sun et al., 2025, Kim et al., 2025, Wu et al., 2025]. PersonalLLM creates synthetic preference models [Zollo et al., 2024]. Simulators reduce cost but can impose their own style and reward biases [Rezk et al., 2025]; real-user evidence remains necessary.

Prior-work setting. **One user:** simulated personas, not a persistent target policy; offline initialization: method-dependent; post-deployment update: no; personalized agent: several methods; LLM-policy training: method-dependent. **Difference:** simulated users cannot establish improvement for one deployed real user.

P11: make per-user deployment cheaper than repeated large-model use. Per-user adapters and online training add storage and compute. OPPU uses lightweight adapters [Tan et al., 2024], CoSteer steers a cloud model from a local model [Lv et al., 2025], and FABLE learns a small external policy [Jin et al., 2026]. We test whether a compact user policy can match a larger generic model at lower serving cost.

Prior-work setting. **One user:** yes; offline initialization: OPPU only; post-deployment update: FABLE only; personalized agent: FABLE only; LLM-policy training: OPPU only. **Difference:** CoSteer is tuning-free and FABLE trains an external policy; none covers both phases.

P12: evaluate future behavior rather than history recall. LongLaMP provides chronological long-form tasks [Kumar et al., 2024]; PrefEval tests preference following [Zhao et al., 2025]; PERMA adds evolving preferences and tasks [Liu et al., 2026]; and PRISM supplies individual feedback [Kirk et al., 2024]. Our endpoint is behavior on untouched future tasks.

Prior-work setting. **One user:** represented in the data; offline initialization: not applicable; post-deployment update: no; personalized agent: PERMA only; LLM-policy training: no. **Difference:** these are evaluation resources, not update algorithms.

3 Claim

A compact, user-owned language-model policy can be initialized from one person’s heterogeneous historical interactions and then improved from that person’s later agent-task evidence through real parameter updates, while preserving task success and safety.

This requires an offline personal checkpoint, later parameter updates, and paired evaluation of personalization and task success. The method contribution is the update rule connecting history, fresh trajectories, feedback, replay, and checkpoint activation.

3.1 Additional claims

Report these claims only when their tests pass:

Fast adaptation. Fewer real interactions reach the same valid utility.

Trajectory advantage. Tool actions and verified outcomes improve future execution beyond text-only history.

Small versus large. A compact personalized policy beats a larger generic model while preserving task success and lowering cost.

Simulation benefit. Simulator training reduces real-user interactions and survives held-out human evaluation.

Privacy. The implementation follows its declared data boundary without exposing another user’s raw records.

4 Success Criteria

4.1 Hard validity gates

Before comparing scores, every reported update must pass:

1. the parent and candidate checkpoint hashes differ;
2. the candidate loads in a fresh worker and actually generates the evaluated trajectories;
3. offline, online-adaptation, validation, and future-test intervals are chronological and disjoint;
4. every training record has user, task, behavior-policy, checkpoint, timestamp, and feedback provenance;
5. failed runs, users, and rollbacks remain in the report;
6. the simulator, reward model, or LLM judge has not seen the future-test target or its answer.

Failure of any item invalidates the run rather than lowering its score.

Table 1: Success criteria. The first four rows test the main claim.

Claim	Measurement	Pass condition
Personalized improvement	Paired future-task Δ_{pers} against the strongest baseline.	Lower 95% bound is above zero and the point estimate exceeds the preregistered meaningful effect.
Task preservation	Paired task success or environment reward.	Lower 95% bound exceeds $-\delta_{\text{task}}$.
Value of online learning	Candidate versus frozen offline checkpoint and naive cumulative update.	Candidate passes equation (5) against both.
Real policy change	Parameter count, hashes, load receipt, and changed outputs.	Nonzero finite update; candidate is loaded for evaluation.
Fast adaptation	Area under the future-utility learning curve and interactions-to-target.	Higher clustered 95% interval for area, or fewer interactions to the same preregistered target.
Retention	Backward transfer on earlier stable preferences and tasks.	No task-margin violation; less forgetting than naive continual training.
Drift recovery	Utility after a controlled persistent preference change.	Faster recovery without elevated updates on transient-noise controls.
Safety	Severe violations and policy regressions under a fixed panel.	No increase in severe events; lower-severity change stays within its preregistered margin.
Small versus large	Compact personalized model versus large generic, RAG, and profile-enabled models under equal information.	Personalization superiority, task non-inferiority, and strictly lower serving cost all hold.
Simulator benefit	Human-simulator agreement, calibration, and method ranking.	Simulator-trained gain survives held-out human evaluation; simulation is never the only final judge.
Efficiency	Training GPU-hours, inference latency, memory, storage, tokens, API cost, and user-feedback count.	Candidate lies on the quality-cost Pareto frontier under matched budgets.
Ownership/privacy	Raw-record flow, communication bytes, cross-user leakage probes.	Declared boundary is met with no unauthorized raw-history transfer.

4.2 Primary statistical gate

Let b be the strongest eligible baseline on the validation set. For each user-period cluster, evaluate the candidate and b on identical future tasks. Define

$$\Delta_{\text{pers}} = U_{\text{pers}}(c_{\text{candidate}}) - U_{\text{pers}}(c_b), \quad (3)$$

$$\Delta_{\text{task}} = U_{\text{task}}(c_{\text{candidate}}) - U_{\text{task}}(c_b). \quad (4)$$

The main claim passes only when

$$\text{LCB}_{95\%}(\Delta_{\text{pers}}) > 0 \quad \text{and} \quad \text{LCB}_{95\%}(\Delta_{\text{task}}) \geq -\delta_{\text{task}}. \quad (5)$$

The minimum meaningful personalization effect and δ_{task} must be fixed before final evaluation. Confidence intervals are clustered by user or task lineage, not by treating correlated turns as independent samples.

4.3 Large-model comparison

The large-model comparison must include:

Table 2: Minimum baseline families.

Family	Purpose	Representative implementation
No personalization	Measures the value of any user information.	Same-size base and large API model.
Prompt/profile/RAG	Tests whether parameter updates are necessary.	Full-history prompting, retrieval, and generated profile.
Offline per-user PEFT	Tests the value of the online phase.	OPPU [Tan et al., 2024].
Shared personalized policy	Tests one-user ownership against a population model.	P-RLHF, CoPL, or PROPER [Li et al., 2024, Choi et al., 2025, Zhang et al., 2025].
Continual per-user update	Tests the proposed transition and replay rule.	SPRInG and naive cumulative SFT [Kim and Kim, 2026].
Online-only	Measures the value of offline initialization.	Same optimizer initialized from the generic checkpoint.
Offline then naive online	Isolates the new coupling mechanism.	Sequential SFT/DPO/RL with no special replay or activation rule.
Weight-free online control	Tests whether policy training is worth its cost.	FABLE and CoSteer [Jin et al., 2026, Lv et al., 2025].
Large-model teacher	Tests compact-model competitiveness.	The CLI/API model that produced the offline trajectories.

1. a large generic model without personal history;
2. the same model with the full allowed history;
3. strong RAG or profile prompting;
4. a decoding-time method such as CoSteer when executable [Lv et al., 2025];
5. the compact offline-only and compact online-adapted policies.

Match prompts, tools, history, and task budgets. A compact model wins only if it improves personalization, preserves task success, and costs less.

4.4 Adaptation speed

For threshold τ , measure real interactions rather than epochs:

$$T_\tau = \min\{n : U_{\text{pers}}(c_u^{(n)}) \geq \tau \text{ and task/safety gates pass}\}. \quad (6)$$

Also report learning-curve area, final utility, clarifications, correction time, and rollbacks.

4.5 Simulator evidence

Test:

1. **Preference fidelity:** agreement and calibration on held-out real-user comparisons.
2. **Method fidelity:** rank correlation between methods under the simulator and under real-user evaluation.
3. **Transfer value:** fewer real interactions are needed after simulator pretraining than without it.

Simulator reward alone does not support a real-user claim.

5 Experimental Comparison Set

Evaluation substrates. Use a chronological stream such as LongLaMP [Kumar et al., 2024] and an executable agent benchmark such as PersonalWAB or FingerTip 20K [Cai et al., 2025, Yang et al., 2026].

PrefEval and PERMA add long-horizon tests [Zhao et al., 2025, Liu et al., 2026]. Final results need held-out human feedback or frozen real-user records and two distinct substrates.

Replication. Repeat the protocol across independent users. Report per-user results, the median, worst decile, and cluster-aware intervals, with at least three seeds for feasibility and five for final results.

6 Claims We Must Not Make

- Chronological offline replay is not post-deployment online learning.
- Better retrieval or profile recall is not evidence of a better policy.
- A simulator-only win is not evidence of real-user improvement.
- A changed loss value is not evidence that the candidate checkpoint was loaded.
- A personalized reward-model gain is not automatically a policy or task gain [Rezk et al., 2025].
- Local adapters do not by themselves prove privacy.
- Beating a large model on style does not mean beating it on task capability.
- Mean improvement cannot hide severe failures for a subset of users.

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